

Name: _____

ALGEBRAIC EXPRESSIONS, (EXPANDING DOUBLE BRACKETS)

Date: _____

LO: I can expand the product of two binomials using the FOIL method or grid method.

What does expanding double brackets mean?

Expanding double brackets means multiplying two expressions like $(x + 3)(x + 5)$ to remove the brackets and write as a single expression.

FOIL: First, Outside, Inside, Last. Multiply each pair then collect like terms.

Quick examples

●	$(x + 2)(x + 3) = x^2 + 5x + 6$	FOIL
●	$(x + 4)(x - 1) = x^2 + 3x - 4$	mixed signs
●	$(x - 3)(x - 2) = x^2 - 5x + 6$	both negative
●	$(x + 2)(x + 3) = x^2 + 6$	forgot middle term
●	$(x + 4)^2 = x^2 + 16$	must use FOIL

Key idea

Multiply each term in the first bracket by each term in the second bracket, then simplify by collecting like terms.

$$(x + a)(x + b) = x^2 + (a + b)x + ab$$

Middle term = sum, last term = product

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - both positive

Expand $(x + 2)(x + 5)$

F: $x \times x = x^2$, O: $x \times 5 = 5x$, I: $2 \times x = 2x$, L: $2 \times 5 = 10$

Combine: $x^2 + 5x + 2x + 10$

Simplify: $x^2 + 7x + 10$

$$= x^2 + 7x + 10$$

Collect the two middle terms: $5x + 2x = 7x$.

Example 2 - one negative

Expand $(x + 4)(x - 3)$

F: x^2 , O: $-3x$, I: $4x$, L: $4 \times (-3) = -12$

Combine: $x^2 - 3x + 4x - 12$

Simplify: $x^2 + x - 12$

$$= x^2 + x - 12$$

Be careful with signs: $+4 \times -3 = -12$.

Example 3 - both negative

Expand $(x - 3)(x - 4)$

F: x^2 , O: $-4x$, I: $-3x$, L: $(-3)(-4) = +12$

Combine: $x^2 - 4x - 3x + 12$

Simplify: $x^2 - 7x + 12$

$$= x^2 - 7x + 12$$

Negative $\times$ negative = positive for the last term.

Example 4 - perfect square

Expand $(x + 5)^2$

Write as $(x + 5)(x + 5)$

F: x^2 , O: $5x$, I: $5x$, L: 25

Simplify: $x^2 + 10x + 25$

$$= x^2 + 10x + 25$$

A perfect square always gives $x^2 + 2ax + a^2$.

YOUR TURN – PRACTICE (deliberate practice)

1. Both terms positive

Expand and simplify.

a) $(x + 1)(x + 3)$

b) $(x + 2)(x + 4)$

c) $(x + 5)(x + 2)$

d) $(x + 6)(x + 1)$

2. One negative term

Expand and simplify. Watch the signs!

a) $(x + 5)(x - 2)$

b) $(x + 3)(x - 4)$

c) $(x - 1)(x + 6)$

d) $(x + 7)(x - 3)$

3. Both terms negative

Expand and simplify.

a) $(x - 2)(x - 3)$

b) $(x - 5)(x - 1)$

c) $(x - 4)(x - 4)$

d) $(x - 6)(x - 2)$

4. Applied / special cases

Expand and simplify.

a) $(x + 3)^2$

b) $(x - 4)^2$

c) $(x + 2)(x - 2)$

d) $(2x + 1)(x + 3)$

5. Mixed expansion

Expand and simplify each expression.

a) $(x + 4)(x - 5)$

b) $(x - 7)(x + 2)$

c) $(x + 1)^2$

d) $(x - 3)(x + 3)$

e) $(2x + 3)(x - 1)$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $(x + 2)(x + 3) = x^2 + 5x + 6$ ☐

Correct answer: _____

Reason: _____

b) $(x + 4)^2 = x^2 + 16$ ☐

Correct answer: _____

Reason: _____

c) $(x - 3)(x - 2) = x^2 - 5x + 6$ ☐

Correct answer: _____

Reason: _____

d) $(x + 5)(x - 5) = x^2 + 25$ ☐

Correct answer: _____

Reason: _____

e) $(x + a)(x + b)$ has middle term $(a + b)x$ ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A rectangle has sides $(x + 3)$ cm and $(x + 7)$ cm. Write an expression for the area. If $x = 5$, find the numerical area and verify by substituting into your expanded form.

Expression: _____

Simplified: _____

b) Show that $(x + a)(x - a) = x^2 - a^2$ for any value of a . Use this result to calculate 103×97 without a calculator (hint: let $x = 100$, $a = 3$).

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

